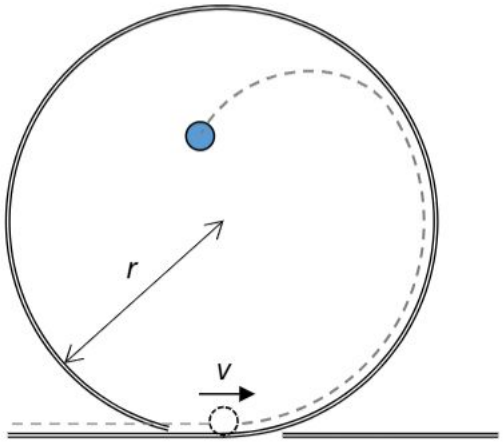
10 A body enters the vertical circular track of radius r shown below with speed v.



If the body falls off the track as shown above, what is a possible value of v?

A $\sqrt{gr}$ **B** $\sqrt{2gr}$ **C** $\sqrt{4gr}$ **D** $\sqrt{6gr}$

Ans: **C**

Min speed at mid-height = 0

Min v to reach mid-height = $\sqrt{2gr}$ since $\frac{1}{2}mv^2 - 0 = mgr$ (conservation of energy)

Min speed at top = $\sqrt{gr}$ since $mg < m\frac{v^2}{r}$ (N2L)

Min v to reach top = $\sqrt{5gr}$ since $\frac{1}{2}mv^2 - \frac{1}{2}mgr = mg(2r)$ (conservation of energy)

Since sphere fell off between mid-height and top, $\sqrt{2gr} < v < \sqrt{5gr}$.